

FPV Sim to VBS4: DIS interoperability brief

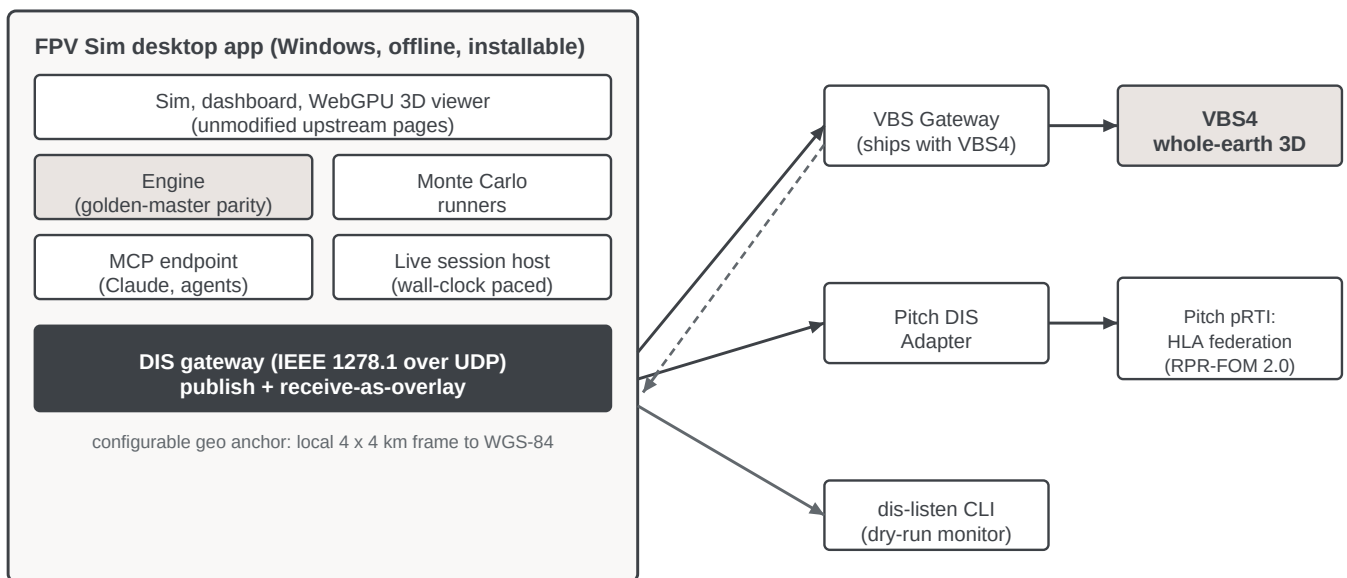
Personal project by Wes Fine. Not an official BAE Systems OneArc publication; VBS4, VBS Gateway and the Pitch products are referenced as interoperability targets. All simulation data is notional and unclassified.

What this is

FPV Sim models a force-on-force engagement: two sides each field armed FPV quadcopters, passive RF direction-finding nodes, and a ground control station (the target). The sides hunt each other's GCS by its RF emissions, so emission discipline (EMCON) decides the fight. The model is deterministic by construction (the same seed always replays the identical engagement), which is what makes its Monte Carlo evidence and its golden-master verification chain possible. It now ships as an installable, fully offline Windows application that runs everything locally: the interactive sim, the studies and dashboard, an MCP endpoint for agent control, and a native DIS gateway that streams live engagements to external simulators.

The demo: start a live, wall-clock-paced engagement in the app (from the UI, or by asking Claude through MCP), and VBS4, connected over DIS through its standard VBS Gateway, renders the airframes, ground stations, emissions and strikes on correlated terrain. External DIS entities appear back in the app as a read-only overlay that can never influence the engagement.

Architecture



Outbound DIS: Entity State, EM Emission, Detonation, Fire (optional), Start/Resume, Stop/Freeze.

Inbound (dashed): external Entity State PDUs render as a read-only overlay; nothing from the network reaches the engine.

What goes on the wire

PDU	When it goes out	Key choices
Entity State	first sighting, dead-reckoning threshold trips (2 m or 3 degrees against a DRM mirror), 5 s heartbeat, appearance changes, final state at death	statics DRM 1, drones DRM 2 (FPW) by default. Wrecks keep heartbeating with destroyed appearance (plus a 60 s flaming window on kills) until session stop. Markings: B-GCS, O-sUAS-3
EM Emission	every keyed to unkeyed edge of the GCS C2 uplink and each drone video downlink, 10 s heartbeat while keyed, zero-beam EE at unkey	one omni beam per emitter. RF values are notional config (uplink 915 MHz, video 5.8 GHz): the engine models keying, not a link budget. The EMCON duty cycles that decide the fight are visible on the wire
Detonation	the tick a one-way FPV impacts	GCS kill: Entity Impact with the target ID. Objective strike: Ground Impact. The airframe is the munition. Order: killer's final ESPDU, then Detonation, then the target's destroyed ESPDU
Fire	off by default (config)	a one-way FPV has no separate launch event, the Detonation stands alone. Enable for consumers that pair Fire with Detonation
Start/Resume, Stop/Freeze	session start announce, pause (Recess) and resume; stop sends nothing unless the end-VBS-mission option is set (Termination)	Real-World Time zero, meaning immediate action, which VBS Gateway honors. Received sim-management PDUs are logged, never obeyed

Protocol version 6 by default (7 available), receive accepts 4 to 7. Timestamps relative by default. Worst-case rate at real time: roughly 25 to 30 PDUs per second, under 5 kB/s.

Time across the DIS boundary

DIS has no time-scale message, so no peer can be commanded to a speed, and none needs to be. At accelerated session speeds the gateway publishes wall-apparent kinematics: velocity and angular-rate fields scale by the speed factor (attitude stays physical), and every speed change fires a full Entity State refresh volley. Receivers extrapolate in wall time, so VBS4 renders the engagement at the app's chosen speed with correct smoothing, its own clock untouched. Pause and resume travel as Stop/Freeze (Recess) and Start/Resume with immediate-action semantics. An advisory ceiling of 8x mirrors VBS's own documented smoothing limits under accelerated time.

Terrain and georeferencing

A configured anchor (latitude, longitude, height of sim sea level, grid azimuth) places the sim's 4 x 4 km procedural terrain on Earth; entity positions and the DEM export share one mapping, so they cannot disagree. The terrain-export tool writes any seed's heightfield as a georeferenced Esri ASCII Grid with a UTM projection file, plus a canopy-density raster, for import through VBS Geo. Set the VBS water level to elevation 0: negative posts are real seabed, and the coastline then renders where the engagement thinks it is.

Two guarantees worth knowing

Determinism: a live session that runs to its end produces a result byte-identical to the batch run of the same seed, mode and configuration (enforced in CI on every commit). Isolation: received DIS traffic feeds a display-only overlay; there is no code path from the network into the engine.

Lab procedure (condensed from docs/VBS4_CHECKLIST.md)

0	Dry run without VBS. npm run dis-listen -- --port=3000 --mode=broadcast on any second machine. Expect one Start/Resume, six ESPDUs immediately then drones on launch, EE lines flipping with the EMCON schedule, exactly one Detonation at the kill, no malformed lines. Wireshark's DIS dissector should show a clean capture.
1	VBS Gateway configuration. DIS adapter only (never DIS and HLA adapters together, a documented VBS limitation). Match DIS version (6), exercise ID (1), UDP port (3000), and the broadcast or multicast scheme. Geofilter off or enclosing the anchor box.
2	Terrain correlation (the better demo). npm run terrain-export -- --seed=N --lat=.. --lon=.. and import the elevation .asc (with .prj) via VBS Geo at the same coordinates. Set water level to elevation 0 (negative posts are real seabed). Verify three grid references before flying.
3	Entity mapping. confirm VBS Gateway resolves the four enumerations to visible models (defaults are generic surrogates chosen for fuzzy mapping), remap per site on either side, record the mapping.
4	Live engagement at 1x. six entities at correct grid refs, then eight after launches. Drones smooth (tune thresholds or switch DRM 2 to 4 if rubber-banding). EE systems listed in the Gateway UI while keyed. Detonation effect at the kill, wreck persists. Pause and resume from the app freezes and resumes VBS-side entities. Then a 4x pass: entities 4x faster, still smooth.
5	Reverse path. place a VBS4 entity inside the box: it appears in the app's Live Ops overlay within about 5 s, extrapolates smoothly, expires about 12 s after deletion. The engagement itself is untouched (the overlay is display-only by construction).
6	Capture tuned values. feed threshold or enumeration adjustments back into the config defaults and archive a reference capture.

Versions and links

Application	fpv-sim-app v0.1.0, installer fpv-sim-app-setup-0.1.0.exe (unsigned, SmartScreen warns). github.com/wasomma/fpv-sim-app (private): docs/INTEROP.md, docs/VBS4_CHECKLIST.md
Simulation	fpv-sim v1.5.0 (UI pin 7050617): the engagement model, Monte Carlo evidence, dashboard, 3D viewer. github.com/wasomma/fpv-sim
Engine	fpv-sim-mcp v0.3.0 (pin f848528): the golden-master verified TypeScript port plus MCP tools. github.com/wasomma/fpv-sim-mcp
Verification	58 automated tests plus an 8-step in-app self-check (including a DIS loopback session) on every commit, and against the packaged build. A live session run to its end reproduces the batch result byte for byte.

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